

# AI Essentials for Professionals

Course Outline | 2026

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## Course Overview

Duration: 20 hours (10 × 2-hour lessons)

Level: Beginner → Confident Professional User

Prerequisites: Basic computer literacy (email, documents, web browsing). No coding required.

## Core Philosophy

*This course develops AI competence, not AI dependency.*

Participants learn how to select appropriate AI tools for common workplace tasks, verify results, and apply AI safely with practical guardrails for privacy and risk.

## Assumed Learner

Adults from a variety of educational backgrounds (high school through university), with little to no prior AI experience. Primary audience includes administrative staff, general office staff, and HR, but the course also works for individual professionals and small business owners.

## Course Learning Outcomes

By the end of the course, participants will be able to:

- **CLO1:** Explain how generative AI works (finding patterns in training data to produce probable outputs) and what it does not do (understand meaning, exercise judgment, or recognise its own errors).
- **CLO2:** Write structured AI prompts using a consistent framework (role, goal, specific details, rules/constraints) and adapt this approach across different task types.
- **CLO3:** Improve everyday professional writing (emails, letters, announcements) for clarity, tone, concision, and correctness while maintaining human judgment and voice.
- **CLO4:** Convert unstructured inputs (meeting notes, photos, voice memos) into accurate, structured outputs by applying the No-New-Facts rule and verifying AI-generated content.
- **CLO5:** Apply privacy and responsible-use practices throughout AI-assisted work, including data classification, redaction, bias recognition, and The Four Checks framework.
- **CLO6:** Use AI with spreadsheets (Excel/Google Sheets compatible) to clean data, generate formulas, and produce visualizations, with manual verification of outputs.
- **CLO7:** Summarize and extract meaning (themes, risks, action items) from documents using structured prompts, and critically assess AI-generated summaries for completeness and accuracy.
- **CLO8:** Use grounded research tools to produce short, traceable briefs that include cited sources, uncertainty notes, and next-step questions.
- **CLO9:** Produce AI-assisted visual outputs and presentations responsibly, applying awareness of copyright, intellectual property, and cultural/professional appropriation.
- **CLO10:** Use AI as a second opinion for high-stakes text (policies and contract-like documents) by identifying questions, ambiguities, and risks — without relying on AI as an authority.
- **CLO11:** Verify AI outputs consistently across task types by applying lesson-specific verification checkpoints and a personal verification checklist.

## Course Learning Outcome Alignment Grid

● Emphasized in this lesson    ○ Touched on / supporting role    (blank) Not present

| Course Learning Outcome      | Lesson |   |   |   |   |   |   |   |   |    |
|------------------------------|--------|---|---|---|---|---|---|---|---|----|
|                              | 1      | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| CLO1. How AI works           | ●      | ○ |   | ○ |   |   | ○ |   |   | ○  |
| CLO2. Prompt framework       |        | ● | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ●  |
| CLO3. Professional writing   |        | ● | ○ |   |   |   |   |   | ○ | ○  |
| CLO4. Meeting capture        |        |   | ● |   |   |   |   |   |   | ○  |
| CLO5. Safe & responsible use | ○      |   | ○ | ● |   | ○ |   | ○ |   | ●  |
| CLO6. Data & spreadsheets    |        |   |   |   | ● |   |   |   |   | ○  |
| CLO7. Document synthesis     |        |   |   |   |   | ● |   |   | ○ | ○  |
| CLO8. Grounded research      |        |   |   |   |   |   | ● |   | ○ | ○  |
| CLO9. Visuals & presentation | ○      |   |   |   | ○ |   |   | ● |   | ○  |
| CLO10. High-stakes text      |        |   |   |   |   | ○ |   |   | ● | ○  |
| CLO11. Verification          |        | ● | ● | ○ | ● | ● | ● | ○ | ● | ●  |

## Complete Course Outline

### Lesson 1: AI Landscape & Tool Survey

**Focus:** Overview of AI types and hands-on exploration of a wide range of tools.

#### Topics

- How generative AI works: pattern matching, not understanding
- AI vs Generative AI vs embedded AI features
- Five tool categories: chat & generate, search-grounded, document Q&A, capture-to-structure, creative/visual
- Key AI risks: accuracy, bias, privacy, security
- Human in the loop
- Account setup and basic usage

#### Learning Goals

By the end of this lesson, participants will be able to:

- Explain in plain language how generative AI produces output — finding patterns in training data to generate probable responses — and describe at least two situations where this makes AI output unreliable regardless of how confident it sounds.
- Describe the distinction between AI, Generative AI, and AI features embedded in everyday software, giving one workplace example of each.
- Classify AI tools into five functional categories and describe the primary use case and a key limitation of each.
- Explain the key risks of AI tools: accuracy (hallucinations), potential bias, privacy considerations, and security concerns.

- Complete a guided tool survey by testing tools across the five categories with the same safe inputs and recording observations.
- Construct a personal Tool Map identifying what each tested tool is best for, where it falls short, and what data should never be shared with it.

### Activity

- Tool Survey: complete a guided checklist across five tool categories using the same safe inputs
- Create a personal Tool Map (what each tool is good for, where it fails, what data not to share)

### Tools Featured

- Chat & Generate (input: question or task; e.g., ChatGPT, Gemini, Claude)\*
- Search-grounded (input: topic to research; e.g., Perplexity + \*)
- Document Q&A (input: document; e.g., NotebookLM + \*)
- Capture-to-structure (input: raw notes or audio; e.g., Notion AI, Microsoft Teams + \*)
- Creative/Visual (input: brief or concept; e.g., Canva, DALL-E, Suno, Gamma)

## Lesson 2: Prompt Engineering & Professional Writing

**Focus:** Using AI to improve professional writing quality while maintaining human judgment and voice.

### Topics

- How LLMs work
- The 5 Cs: Clear, Concise, Correct, Complete, Considerate
- Prompt Engineering framework: role, goal, specific details, rules/constraints
- Constraint-based prompting: what to change, keep, avoid
- Common AI writing failure patterns
- Verification: facts, names, dates, policy references
- Human in the loop: 5 Cs as an approval checklist

### Learning Goals

By the end of this lesson, participants will be able to:

- Gain a high level understanding of how LLMs work: tokenization, embeddings, transformers.
- Write a structured prompt that assigns a role, states a goal, includes specific context, and sets rules and constraints — and explain why each element improves the output.
- Use a constraint-based prompt to improve at least three short workplace communications against the 5 Cs (Clear, Concise, Correct, Complete, Considerate).
- Identify at least two AI writing failure patterns in a set of AI-edited drafts and explain why each is problematic.
- Justify which AI suggestions to accept or reject for each communication, with reference to intent, audience, and factual accuracy.
- Verify a completed AI-assisted communication by checking all factual elements (names, dates, figures, policy references) before sending.
- Construct a reusable everyday-writing prompt template and a companion QA prompt that can be saved and applied to future tasks.

### Activity

- Warm-up: try an unconstrained prompt first, then a structured prompt — compare and discuss the difference
- Email Clinic: improve 3 short communications using the 5 Cs; document which AI suggestions you accepted, rejected, and why
- Create a reusable 'everyday writing' prompt template + a quick QA prompt

### **Lesson 3: AI as a Notetaker & Operational Writing**

**Focus:** *Turning messy inputs into accurate outputs, including photos/scans and single-person voice memos.*

#### **Topics**

- Capture inputs: handwritten notes, photos/scans, voice memos
- Structured outputs: minutes, decisions, action tables
- No-New-Facts rule
- Data sensitivity preview (see Lesson 4)
- Verification: QA prompt to detect invented details
- Human in the loop: AI captures structure, not authority

#### **Learning Goals**

By the end of this lesson, participants will be able to:

- Convert a messy handwritten or scanned meeting note into a structured set of minutes, a decisions list, and an action table using AI, applying the prompt framework to control the output.
- Apply the No-New-Facts rule by explicitly labelling missing owners, deadlines, or decisions as “Not specified” rather than allowing AI to invent them.
- Detect invented or inferred details in AI-generated minutes by running a QA verification prompt and comparing the output to the original source.
- Identify at least one piece of information in a meeting note that should be treated with caution before sharing with an AI tool, anticipating the safe/responsible use framework in Lesson 4.
- Assess when a single-person voice memo provides sufficient basis for minutes and when additional confirmation from participants is required.

#### **Activity**

- Capture-to-Minutes Pack: produce official minutes + action table + follow-up email from (a) scanned notes and (b) a single-person voice memo — Use the prompt framework from Lesson 2 (role, goal, specific details, rules/constraints) when constructing prompts for this activity.
- Add an ‘Unknowns / needs confirmation’ section and run a QA verification prompt to detect invented details

### **Lesson 4: Privacy and Responsible Use — Practical Guardrails**

**Focus:** *Cautious, workplace-appropriate AI use without slowing productivity.*

#### **Topics**

- Data sensitivity tiers: safe / caution / do-not-share
- Redaction and prompt reframing
- Recognizing bias and incomplete perspectives
- The Four Checks: Before / During / After / Always
- Verification: bias, completeness, and sensitivity checks
- Human in the loop: accountability cannot be delegated

#### **Learning Goals**

By the end of this lesson, participants will be able to:

- Classify workplace data into safe / caution / do-not-share categories and justify each classification with reference to sensitivity and context.

- Reframe at least three risky prompts using redaction and abstraction techniques while preserving the practical usefulness of the AI output.
- Detect signs of AI bias or incomplete perspective in a generated output and generate a follow-up prompt to surface missing viewpoints.
- Apply The Four Checks to any AI-assisted task — Before: Is this safe to share? During: What has the AI assumed that I didn't say? After: Have I checked the facts? Always: This output is mine — I'm responsible for it.

#### Activity

- Triage exercise: decide what can be pasted into AI tools and what must be redacted or avoided — Use the prompt framework from Lesson 2 (role, goal, specific details, rules/constraints) when constructing prompts for this activity.
- Write a safe prompt + verification plan for several short workplace snippets (sanitized examples)

### Lesson 5: Data Analysis & Visualization with AI

**Focus:** *Using AI as a data assistant — no prior spreadsheet expertise required.*

#### Topics

- AI as formula and data assistant (plain-language requests)
- Data cleaning and transformation
- AI-generated charts and graphs
- Plain-language insight summaries
- Verification: manual sanity checks on values, charts, and trends
- Human in the loop: AI suggests, human interprets and confirms

#### Learning Goals

By the end of this lesson, participants will be able to:

- Describe in plain language what data operation is needed and use AI to generate the appropriate formula or transformation, without requiring prior Excel knowledge.
- Use AI to clean a dataset and produce at least one chart from a provided spreadsheet.
- Produce a plain-language insight summary that accurately describes what a chart shows and what it does not show.
- Verify at least one AI-computed value and one trend claim by manually checking the underlying data and identifying any discrepancies.
- Judge the reliability of an AI-generated data analysis by identifying assumptions, limitations, and cases where human review changed the output.

#### Activity

- Formula Assistant warm-up: describe a calculation in plain language, have AI generate the formula, then verify it manually
- Dataset → clean → chart → insight summary: produce at least one chart and a plain-language summary with stated limitations — Use the prompt framework from Lesson 2 (role, goal, specific details, rules/constraints) when constructing prompts for this activity.
- Manual verification: check at least one computed value and one trend claim against the raw data

### Lesson 6: Reviewing & Summarizing Documents — Extract Meaning and Themes

**Focus:** *Document synthesis for professional use: summaries, themes, and structured extraction.*

#### Topics

- Multi-resolution summaries: 30-second, 2-minute, 1-page
- Theme extraction: claims, evidence, risks, action items
- CV/resume screening: consistent criteria and limits of AI judgment
- Legal and ethical considerations in AI-assisted hiring
- Verification: cross-check, spot-check, uncertainty flags
- Human in the loop: AI extracts, human judges; hiring decisions rest with humans

### Learning Goals

By the end of this lesson, participants will be able to:

- Produce summaries of a complex document at three levels of resolution (30-second, 2-minute, and 1-page) using structured AI prompts.
- Extract themes, risk flags, and action items from a document into a structured table, distinguishing between claims supported by evidence and unsupported assertions.
- Assess the completeness and accuracy of an AI-generated document summary by cross-checking key claims against the source.
- Apply a consistent rubric to an AI-assisted CV screening task, documenting gaps and questions for human follow-up rather than making decisions based on AI output alone.
- Explain the key legal and ethical risks of using AI in hiring, including bias amplification, lack of explainability, and the requirement for human decision-making accountability.

### Activity

- Document Synthesis Sprint (group): each group receives the same 1–2 page document; teams write a brief manual summary (5 min), then use AI to produce summaries at three resolutions and extract themes; groups compare and discuss what the AI got right, missed, or distorted — Use the prompt framework from Lesson 2 (role, goal, specific details, rules/constraints) when constructing prompts for this activity.
- CV Screening Discussion: using a sample anonymized CV set and a provided rubric, apply AI to shortlist and flag gaps — followed by facilitated discussion on bias, legal constraints, and human accountability in hiring

## Lesson 7: Research & Grounded Answers — Sources and Traceability

*Focus: Producing reliable briefs with sources and uncertainty notes.*

### Topics

- Chat tools vs grounded research tools: when to use each
- Requesting sources and evaluating credibility
- Uncertainty language and next-step questions
- Defining ‘good enough’ for workplace research
- Verification: source credibility, citation accuracy, unconfirmable claims
- Human in the loop: AI finds and organizes, human evaluates and decides

### Learning Goals

By the end of this lesson, participants will be able to:

- Distinguish between chat-generative and search-grounded AI tools and explain when each is appropriate for a given research task.
- Produce a manager-ready research brief that includes a short recommendation, cited sources, explicit uncertainty notes, and at least two ‘what we don’t know’ next-step questions.
- Compare the output of a grounded research tool and a generative chat tool for the same query, identifying specific differences in accuracy, sourcing, and confidence language.
- Evaluate the credibility of AI-cited sources using a structured checklist and flag any invented or unverifiable citations.

### Activity

- Manager-ready Brief: answer a workplace question with sources, a short recommendation, and a 'what we don't know' section — Use the prompt framework from Lesson 2 (role, goal, specific details, rules/constraints) when constructing prompts for this activity.
- Side-by-side comparison: run the same research query through a grounded tool and a chat tool; identify and document the differences

## Lesson 8: Visualization & Presentation

**Focus:** *Creating visual outputs — images, diagrams, and presentations — with AI assistance.*

### Topics

- AI-assisted image generation: diagrams, logos, photos
- AI-assisted presentation creation and refinement
- Copyright and IP: training data, style mimicry, organizational policies
- Cultural appropriation and Indigenous rights in AI-generated imagery
- Evaluating visuals: accuracy, appropriateness, bias
- Human in the loop: AI generates options, human selects and takes responsibility

### Learning Goals

By the end of this lesson, participants will be able to:

- Generate at least two AI-assisted visual outputs (e.g., a diagram and a presentation slide deck) using structured prompts.
- Evaluate AI-generated visuals for accuracy, audience appropriateness, copyright concerns, and potential cultural bias or appropriation.
- Explain the key IP and copyright risks of using AI-generated images in professional work, including organizational policy considerations.
- Describe why AI image generation tools can reproduce cultural stereotypes and infringe on Indigenous cultural and intellectual property, and apply this awareness when selecting or approving visuals.
- Design a short presentation that combines text, data, and visual elements, using AI assistance at each stage while maintaining editorial control.

### Activity

- Image Generation Lab: generate and critically evaluate AI-created visuals for a workplace scenario — assess each for accuracy, appropriateness, copyright risk, and cultural sensitivity — Use the prompt framework from Lesson 2 (role, goal, specific details, rules/constraints) when constructing prompts for this activity.
- Presentation Sprint: build a short AI-assisted presentation using content from earlier in the course; peer-review for accuracy, visual quality, and responsible use

## Lesson 9: Building Without Coding & Introduction to AI Agents

**Focus:** Moving from "using AI to produce text" to "using AI to build working tools" — and understanding what an AI agent is, how it differs from a chatbot, and where it fits in a professional's toolkit.

### Topics

- From prompting to building: what "no-code AI building" means and when it's the right tool for the job
- Overview of no-code/low-code AI builders (e.g., app builders, automation platforms, custom GPT/assistant builders)

- What makes something an "agent": goals, tools, memory, and multi-step autonomous action vs. single-turn chat
- Agent components in plain terms: instructions, tools/actions, triggers, and guardrails
- Common professional use cases: research assistants, intake/triage bots, workflow automations, internal tools
- Risks specific to agents: compounding errors across steps, unintended actions, over-permissioning, and the need for human checkpoints

### Learning Goals

By the end of this lesson, participants will be able to:

- Explain the difference between a conversational AI interaction and an AI agent, using goals, tools, and autonomy as distinguishing factors.
- Identify a real workplace task suited to a simple no-code AI build (e.g., an internal assistant, automation, or mini-tool) versus one better suited to a standard prompt or human process.
- Build a simple no-code AI tool or agent prototype using a builder platform, defining its instructions, scope, and constraints.
- Evaluate an agent design for risk, identifying where human checkpoints, permission limits, or guardrails are needed before deployment.

### Activity

- **Main lab (build):** using a no-code AI builder platform, participants design and build a simple working tool or agent (e.g., a custom assistant for a recurring workplace task) — defining its role, goal, scope, and constraints using the prompt framework from Lesson 2.
- **Mini-lab (risk review):** participants trade agent designs with a partner and complete a short risk checklist — identifying where the agent could take unintended action, where permissions should be limited, and where a human checkpoint is required before the agent's output is used or acted on.

## Lesson 10: Capstone — Applying AI Essentials Across a Real Workplace Scenario

**Focus:** Working as a team to complete a multi-step AI-assisted professional task, demonstrating integrated and responsible AI use.

### Topics

- Integrated team workflow: research → drafting → visualization → review
- Applying the prompt framework, The Four Checks, and No-New-Facts rule throughout
- Peer critique against a structured rubric
- Personal reflection and next steps
- Human in the loop: every step involves deliberate direction and verification

### Learning Goals

By the end of this lesson, participants will be able to:

- Collaborate as a team to complete a multi-step AI-assisted workplace task (research → draft → visual output → briefing for a manager) using appropriate tools and responsible-use practices.
- Apply the prompt framework, The Four Checks, and No-New-Facts rule consistently across all stages of the team task without prompting from the facilitator.
- Critique another team's AI-assisted output using a structured rubric, identifying at least two strengths and two areas where verification or guardrails were insufficient.



- Develop a personal AI use plan — including preferred tools, a verification checklist, and a responsible-use commitment — that can be applied immediately in their workplace.

### Activity

- Team Capstone Task: working in small groups, complete a realistic workplace scenario (e.g., research a topic, draft a briefing note, produce a supporting visual, and prepare a short presentation for a manager) — each team applies the prompt framework and documents their verification steps throughout
- Team Peer Review: exchange outputs with another team and critique using a structured rubric; present one key finding to the group
- Individual close: each participant drafts their personal AI use plan to take back to their workplace